sumologic (NG-SIEM)

A report by

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INTRODUCTION:

Sumologic(NG-SIEM) is a powerful SIEM platform designed to help security teams detect, investigate, and respond to threats with speed and precision. It provides real-time threat detection, automated alert triage, and contextual insights by ingesting and analyzing data from diverse sources. With built-in machine learning and behavioral analytics, it helps security teams identify anomalies, correlate events, and respond swiftly to incidents. Its scalable architecture supports dynamic environments, making it ideal for hybrid and cloud–first organizations. Sumo Logic's intuitive dashboards and integrated case management streamline investigations, empowering SOC teams to stay ahead of evolving threats.

KEY FEATURES:

1.Cloud-Native Architecture: Built for cloud-first environments, it scales automatically with data volume. Works seamlessly across cloud, and containerized setups.

2. Real-Time Threat Detection: Detects anomalies based on user and entity behavior, enriches alerts with external intel feeds, and reduces noise by grouping related events into high-fidelity "Insights".

3.Automated Alert Triage: Automatically correlates signals across sources to surface meaningful threats. Assigns severity based on context, behavior, and known indicators and filters out false positives and low-priority alerts.

4. Contextual Investigation Tools: Linked signals show related events across time and systems, adding geo-IP, user identity and asset context to investigations.

5. Custom Dashboards & Visualizations: Track MTTR, alert volume, and analyst workload and visualize attack types, sources and frequency. Build views tailored to specific use of cases or roles.

6. Integrated Case Management: Analysts can assign cases, add notes, and update status. Maintains a record of actions taken during investigations and supports team-based workflows and escalation paths.

7. Playbooks & Automation: Connects with orchestration platforms for advanced automation (SOAR Integration) and trigger scripts or integrations (e.g., block IP, disable user).

8. Board Integration Ecosystem: SIEM + UEBA combines traditional SIEM with user and entity behavior analytics. APIs & Connectors support custom integrations and third-party tools and ingests.

Let's understand the sumologic SIEM platform with the following tool tasks.

Task 1: Deployment of Sumologic Agent Collector into system

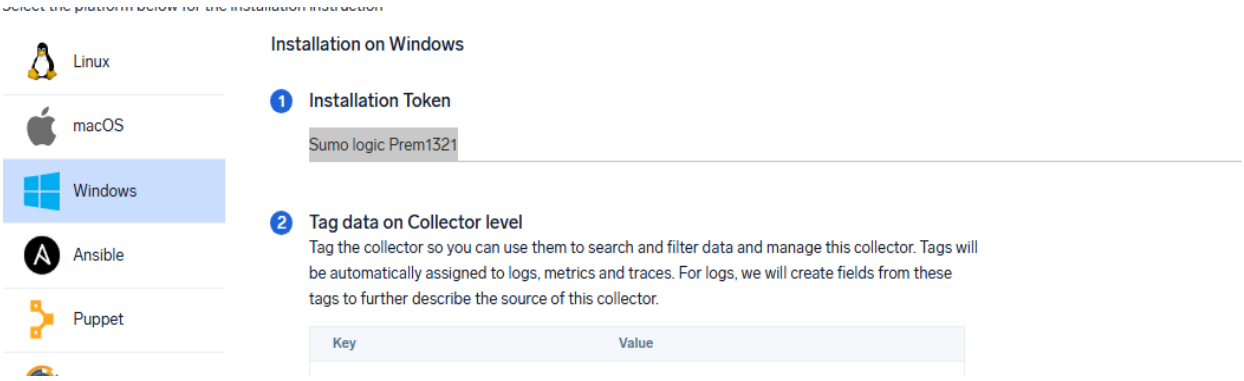
Deploying the Sumo Logic Agent Collector enables secure, real-time log collection from system sources. It compresses, encrypts, and forwards data to sumo Logic for centralized monitoring, analysis, and threat detection.

Step1: Download Agent Collector.

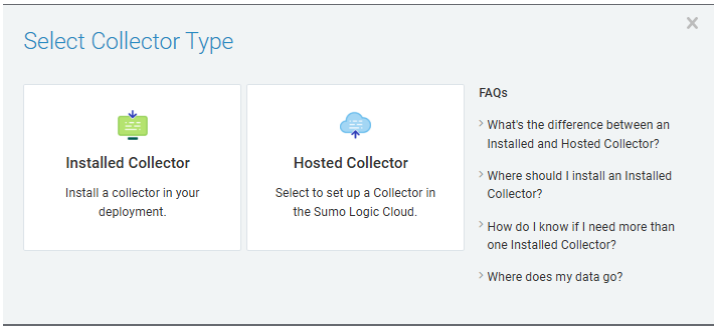
1.Log in to your Sumo Logic account.

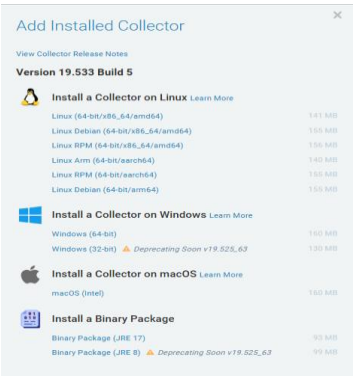
2. Navigate to Data Management > Collection > Add Collector.

3. Create Collector/Installation Token for installing collector used for authentication between Collector and Sumo Logic Cloud.



3. Select Installed Collector > click windows option (based on your operating system).





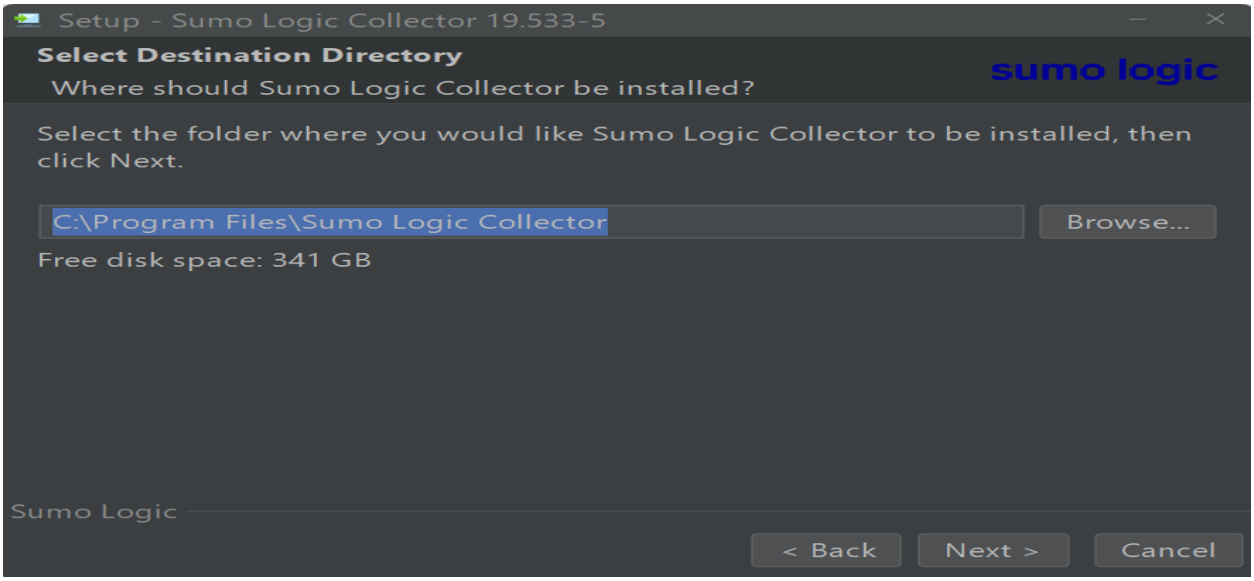
4. Download the Windows Collector Installer(.exe) file.

"C:\Users\premp\Downloads\SumoCollector_windows-x64_19_533-5.exe"

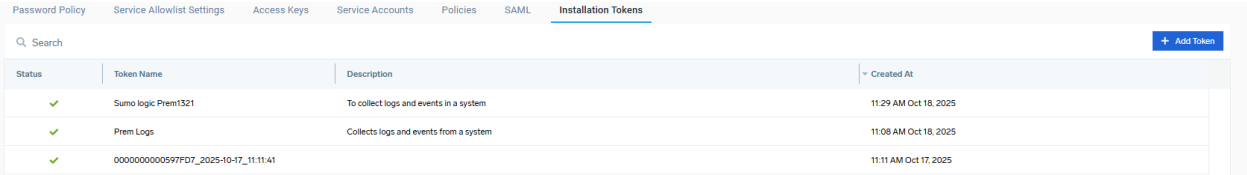
Step2: Install Agent Collector.

1. Run the installer on windows system.

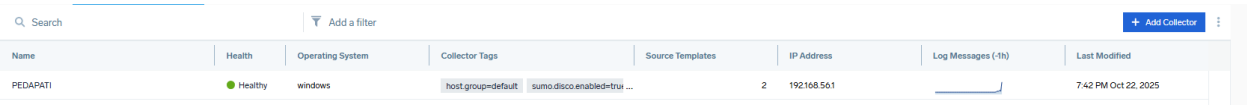
2. During the setup, enter your Collector Name and Installation Path.



3. Paste the Token Code provided by your Sumo Logic account (ensures collector authentication).



4. Complete the installation and verify the collector is visible in the Sumo Logic Console > Manage Data > Collection page.

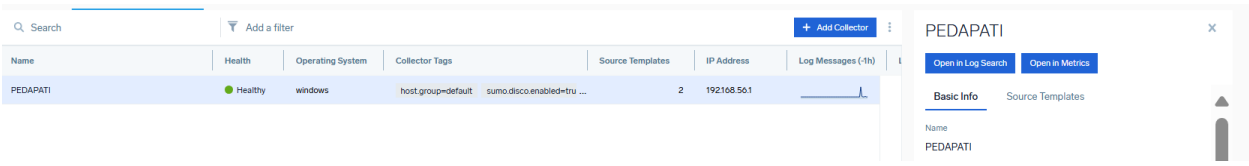


TASK 2: Log Data Collection

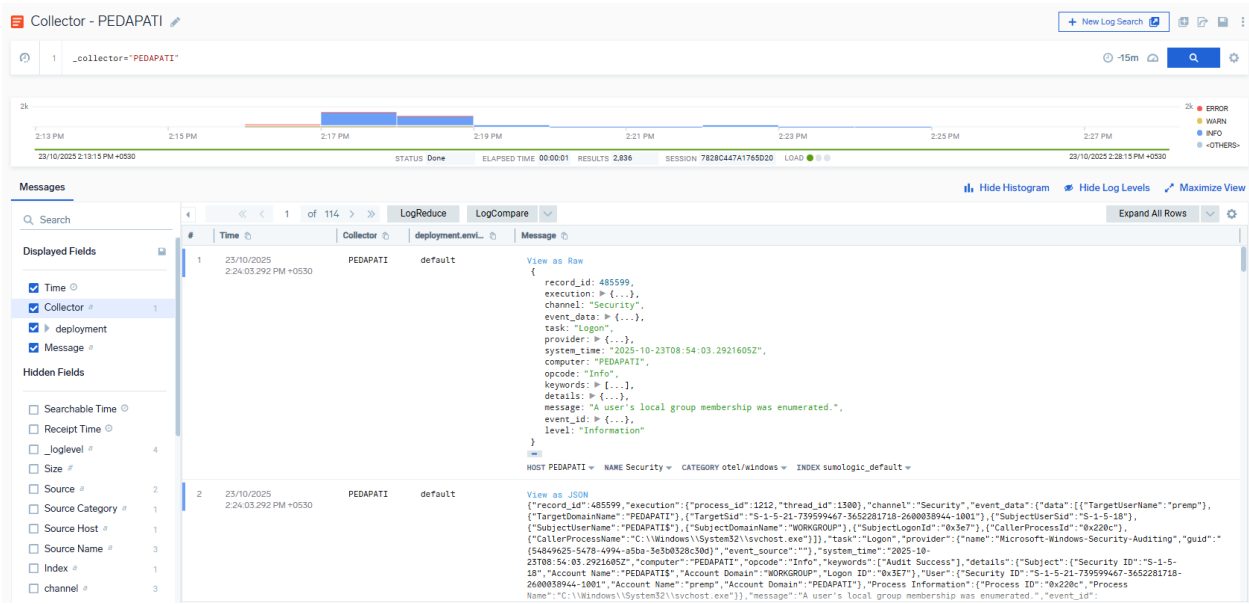
Log data collection involves gathering system, application, and network logs from various sources such as servers, endpoints, and devices. These logs provide critical insights into system performance, user activities, and security events. Centralizing log data enables monitoring, troubleshooting, and detecting anomalies for improved operational visibility and cybersecurity incident response.

1. After installation, open the Sumo Logic Collector Service and collect the log data of the installed machine (windows).

2. Follow the options to collect the log data, open the Sumo Logic Collector Service > Data Management > Collection.



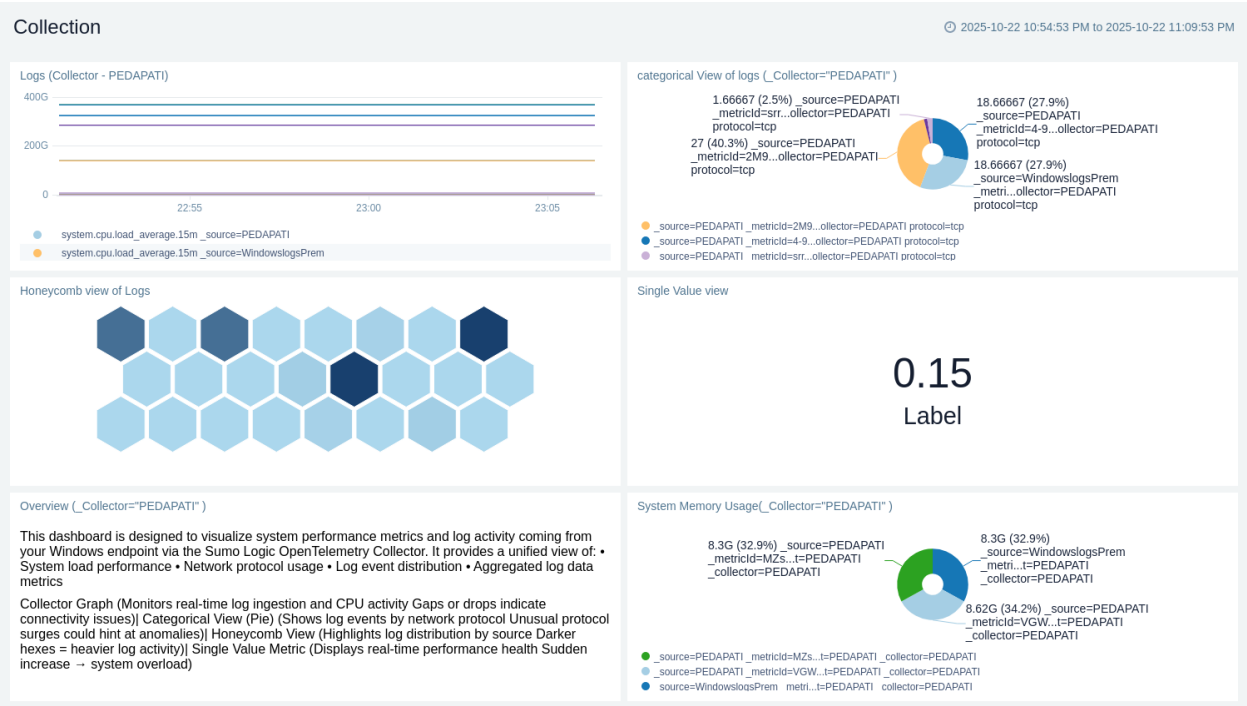
3. Click on > Open in Log Search and collect the logs with queries. (eg.**_collector="PEDAPATI"**).



This image shows a Sumo Logic log analysis dashboard displaying log data from a collector named "PEDAPATI". These logs record Windows Security Events, such as local group membership enumeration – a key indicator of system or user activity monitoring. The log details show various fields inlcuding record_id, event_id, event_data and system.provider.

Task 3: Collect log data to create custom dashboards with visual charts and diagrams.

Sumo Logic SIEM enables real-time log data collection and analysis, empowering users to build custom dashboards with interactive charts and diagrams for threat detection, incident response, and security posture visualization.



Dashboard Overview:

This dashboard is designed to visualize system performance metrics and log activity coming from your Windows endpoint via the Sumo Logic OpenTelemetry Collector

1. Collector Activity Graph (Time-series): A time-series line chart that tracks CPU load average over the past 15 minutes, Log events captured by the Windows system and Confirms that the collector is continuously transmitting metrics to the Sumo Logic cloud.

Query Source: _collector=PEDAPATI and **Metrics** = **system.cpu. load_average.15m**

2. Categorical View (Pie Chart): A categorical (pie chart) visualization showing the percentage distribution of different protocols (e.g., tcp, udp, etc.) being logged or monitored. Provides a network-level overview of what kind of traffic or events are dominant.

Query Source: _collector="PEDAPATI"| aggregate count() by protocol

3. Honeycomb View: A honeycomb heatmap visualization, where each hexagon represents a category or source of log data. Visualizes the density and volume of logs across different sources (Darker hexagons = higher activity or log count).

Query Source: _collector="PEDAPATI" | aggregate count() by _sourceCategory

4. Single Value View: A single numerical indicator summarizing the average CPU load or another key metric (such as average log rate). Provides a quick health snapshot of your system. Lower value (like 0.17) indicates low system load → good performance. Can be used as a KPI (Key Performance Indicator) for monitoring.

Query Source: _collector="PEDAPATI" | aggregate avg(system.cpu.load_average.15m)

5. System Memory Usage (Pie Chart): This type of chart breaks down memory usage distribution among various system sources or metric IDs. Values Observed in Your Dashboard 8.3 GB (32.9%) — _source=PEDAPATI, metric ID linked to system memory zone (MZs),8.3 GB (32.9%) — _source=Windowslogs Prem, same collector — showing memory usage tied to Windows logging services.8.62 GB (34.2%) — _source=PEDAPATI, metric ID linked to VGW (possibly a different monitored instance or host section).

Source Query: collector="PEDAPATI" metric=system.memory.used | aggregate avg(_value) as used_memory by _source, metricId | format("%.2fG", used_memory/1024/1024/1024)

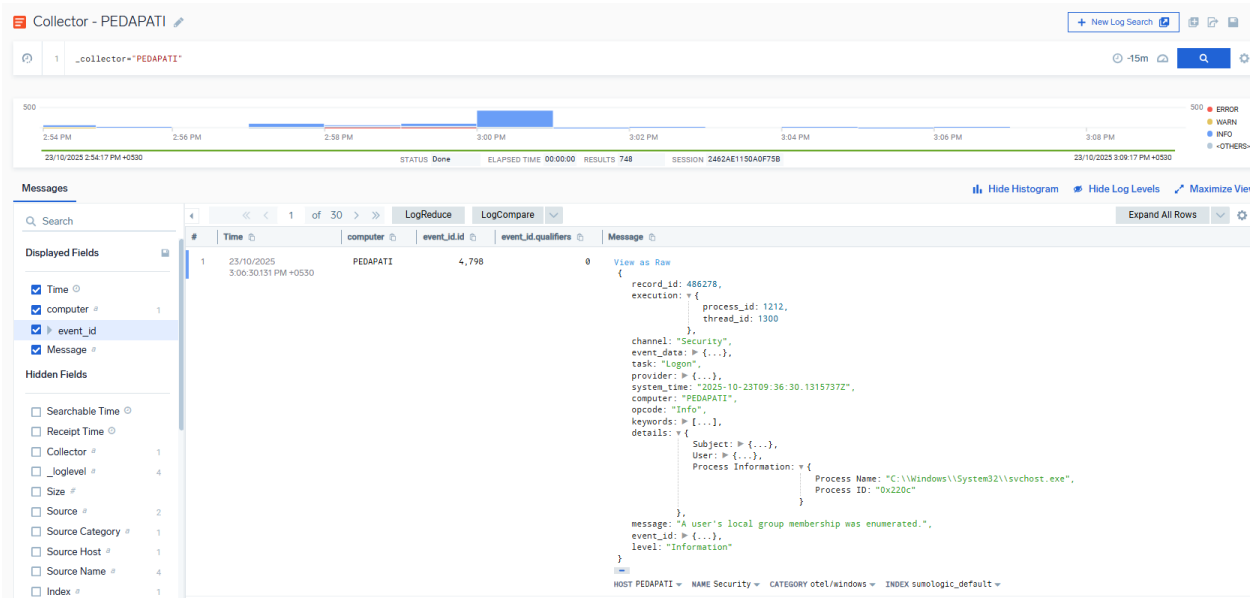
TASK 4: Log Data Analysis & create a detailed report as per your understanding with a screenshot with 3-page report. about tool features.

Log data analysis involves collecting, parsing, and interpreting system, application, and network logs to detect anomalies, troubleshoot issues, and enhance security. It enables security monitoring by detecting intrusions, malware activity, and suspicious behavior across systems and networks. Through performance optimization, it helps identify bottlenecks, system failures, and inefficiencies that could impact operations. It also supports compliance and auditing by providing traceable records that demonstrate adherence to regulatory standards and internal policies.

Let’s analyze the log that was collected through SumoLogic Collector.

Analysis of Log File:

This log entry captures a successful user logon on a Windows system, recorded on 2023-09-26 at 04:36:00 UTC. The event is classified under Event ID 4624, which is commonly used to track authenticated access attempts. The log was generated by the Security Event Log and parsed in JSON format via Sumo Logic.



Findings:

Field	Description
1.record_id	486278 – unique ID for this log record
2.Channel	"Security" – indicates this is a Windows Security Event
3.Task	**"Logon"** – this event is related to a logon or user session
4.Computer	**"PEDAPATI"** – host computer name
5. System_time	**"2025-10-23T09:36:30.1315737Z"** – UTC timestamp
6. Opcode	**"Info"** – log level indicating informational message
7.Eventid	4798 – Windows event code
8.Process Name	**"C:\\Windows\\System32\\svchost.exe"** – process that triggered this event
9. Process_id	**"0x220c"** – internal process identifier
10. Message	"A user's local group membership was enumerated." – indicates user account enumeration

Security Significance: Event ID 4798 (Windows Security Log)

A user's local group membership was enumerated. This means a process (svchost.exe) retrieved information about a user's local group memberships typically done during authentication or administrative queries.